2025-26 S1 – BIO220 PROGRAM

Week	Lecture	Tutorials and Labs	Coursework
Week 1	Introduction to computational Biology.		
Week 2	Charges and Coulomb law. Electrostatic potential.		
Week 3	Atoms and molecules. The Schrodinger equation		
Week 4	Molecular mechanics. Force Fields.		Coursework 1 (in class test): Molecular Physics.
Week 5	Structure of proteins and nucleic acids.	Tutorial and Lab: Molecular visualization with VMD.	
Week 6	Structural models and protein folding. Homology modeling and Alphafold.	Tutorial and Lab: Homology modeling.	Coursework 2 (laboratory): Homology modeling and molecular visualization.
Week 7	Molecular surfaces and electrostatics.		
Week 8	Molecular docking.		Coursework 2 deadline
Week 9	Energy minimization. Normal modes analysis.	Tutorial and Lab: Molecular docking, normal modes.	
Week 10	Free energy and solvation.		
Week 11	Small molecules and chemical spaces. Chemoinformatics.	Tutorial and lab: Protein/Ligand modeling.	Coursework 3 (laboratory): Protein/Ligand modeling.
Week 12	Drug libraries, screening and selection.		
Week 13	Recap.		Coursework 3 deadline

CHARGES AND COULOMB LAW
THE ELECTOSTATIC POTENTIAL

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Xi'an Jiaotong-Liverpool University

西交利物浦大學

Fundamental forces in Nature

Imagine to be a physicist in the 1700-1800 and you have a clear understanding of how gravity work, and where it is originated from. You understand forces and accelerations, and you have a basic understanding of thermodynamics.

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But then you see this:

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Or this…..

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But then you see this:



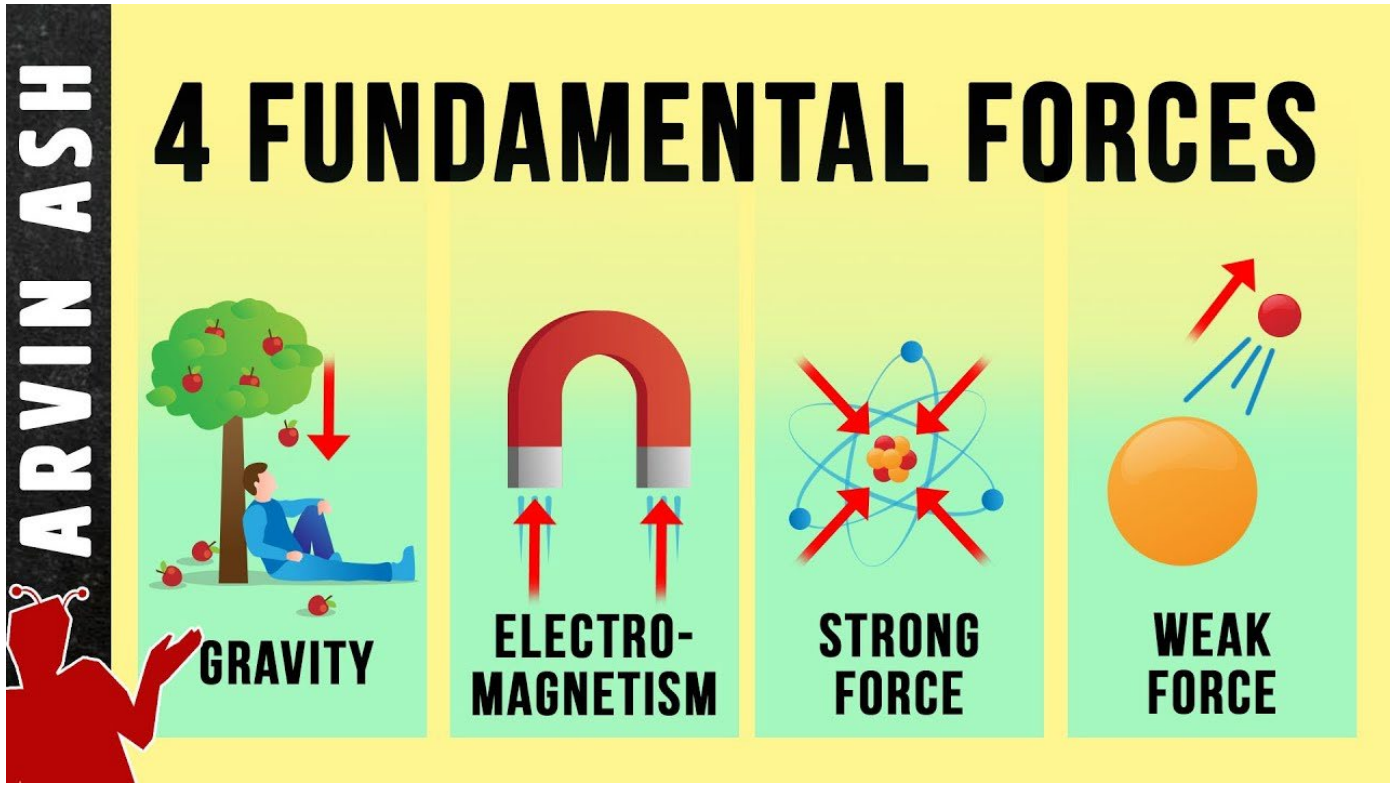
Or this…..

Or this…….



You are sure there are forces you haven't understood, but you don't know where they come from…

Fundamental forces in Nature



Gravity: responsible for attraction of matter having a mass.

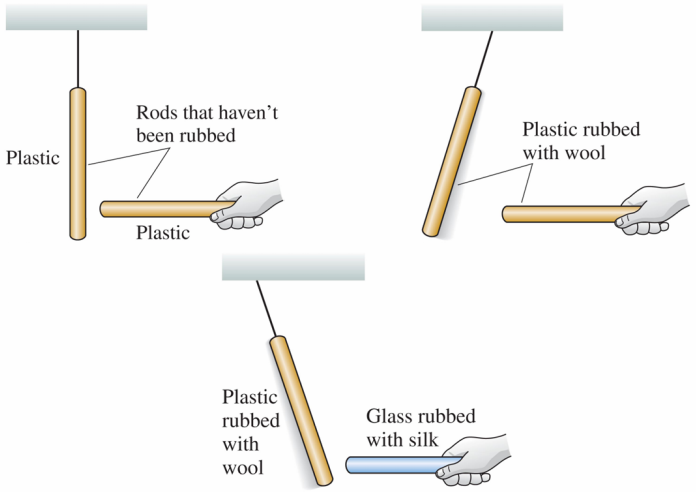
Electromagnetism: interaction between charges, electric and magnetic fields.

Strong force: keeps nuclei together in the atoms.

Weak force: causes beta decay in atoms.

Charges

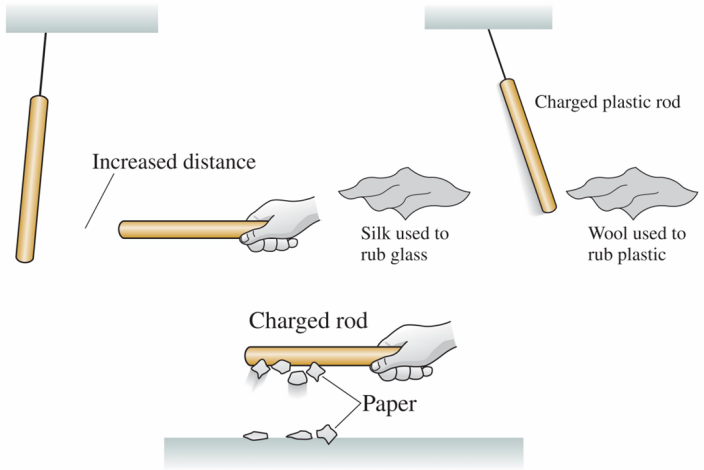
Electric Charges and Forces, Part I



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Slide 20-5

Electric Charges and Forces, Part II



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Experimental evidence show that various objects can be **charged** and interact with forces (without touching).

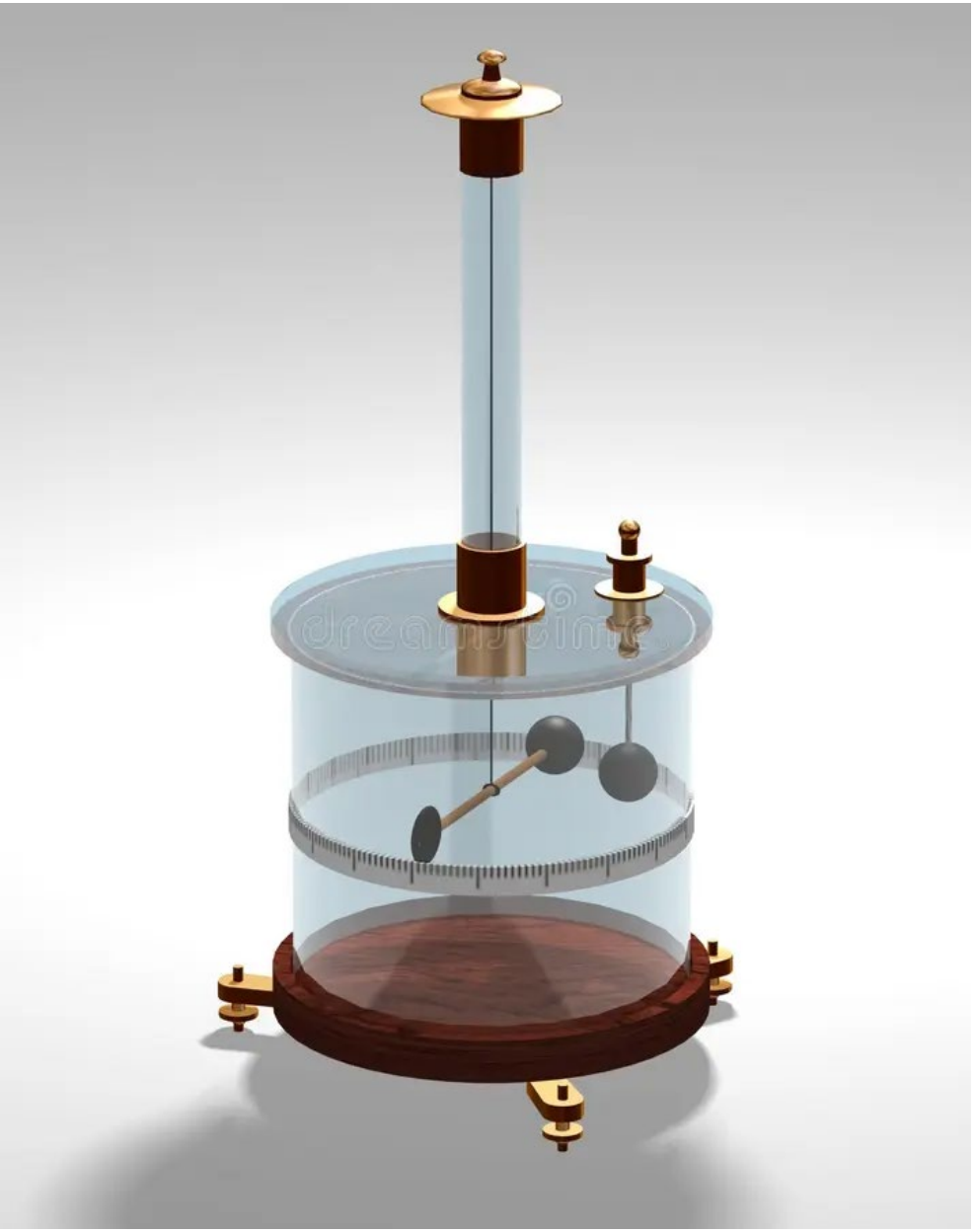
These forces can be stronger than the gravitational force.

Charged objects can attract or repel, indicating the existence of two **opposite** charges (positive or negative).

Similar charged object repel each other, while opposite charged objects attract each other.

The unit of measure of charges is called **Coulomb**, and its symbol is **C**.

Coulomb law



Using a tortional balance, Coulomb was able to carefully measure the force of attraction or repulsion between two charged spheres:

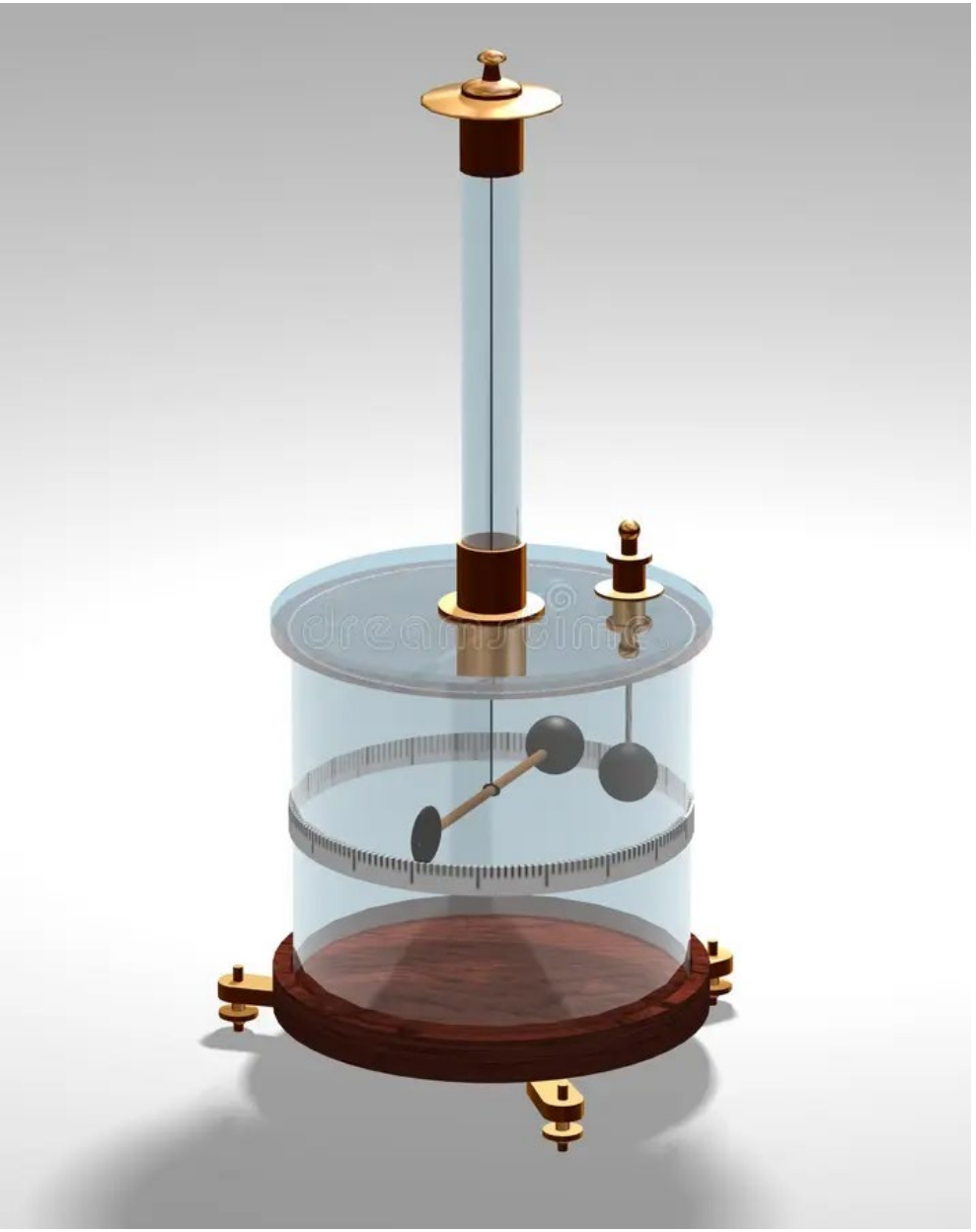
$$F = k\frac{q_1 q_2}{r^2}$$

Where q_1 and q_2 are the charges of the two spheres (in Coulomb), r is their distance (in meters) and $k = 8.99 \cdot 10^9\ Nm^2/C^2$ is a constant.

The constant *k* is often expressed in term of another constant ε_0 which is called also the vacuum permittivity or permittivity of free space:

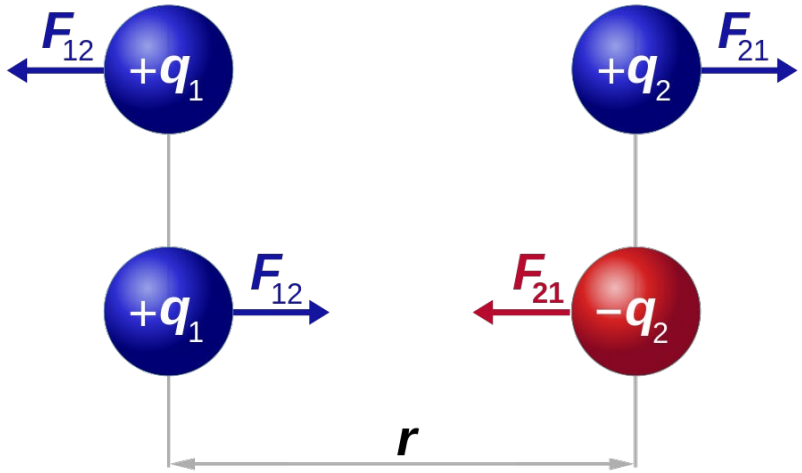
$$k = 1/4\pi\varepsilon_0$$

Coulomb law



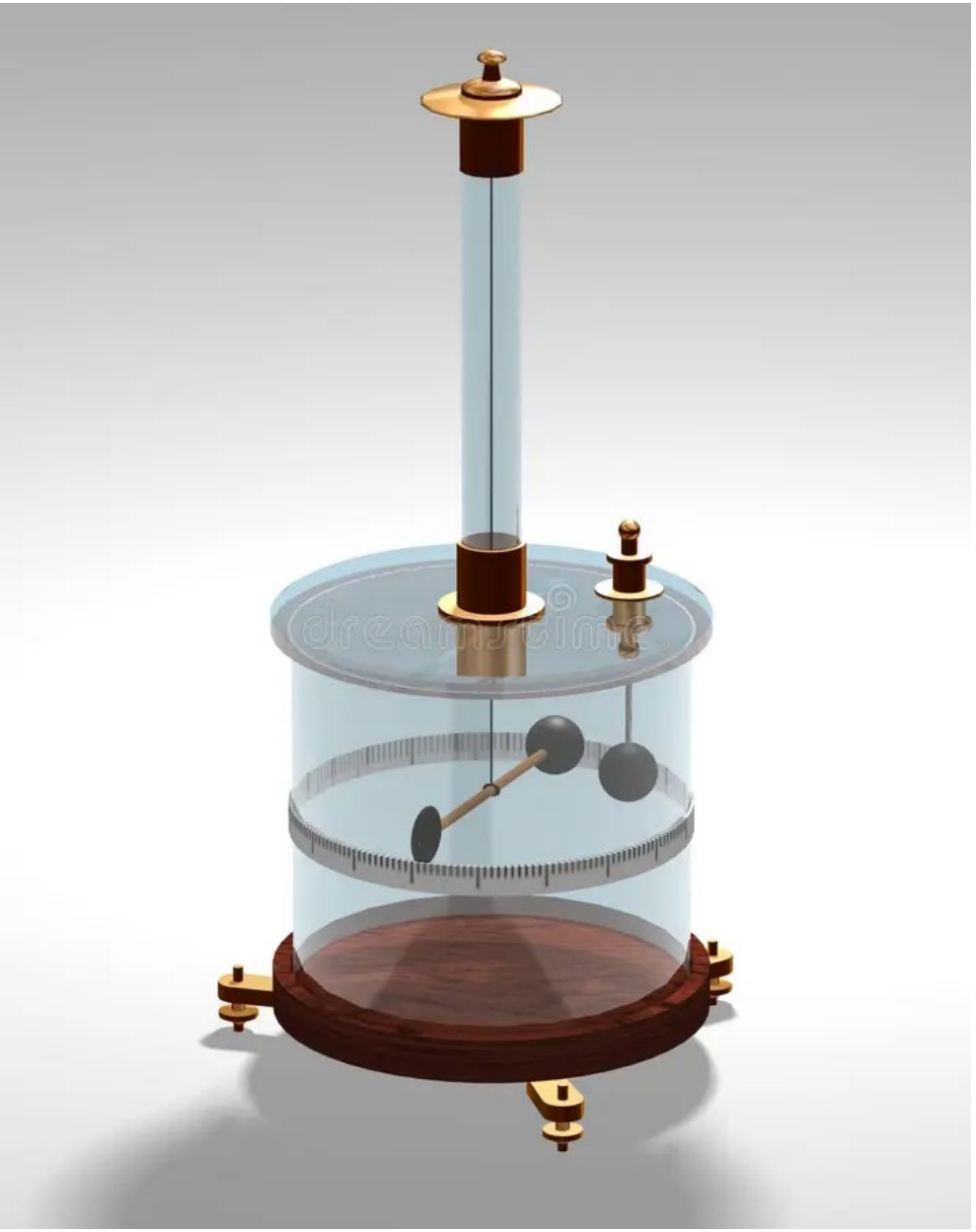
The Coulomb law is similar to Newton's gravitation law. However, masses are only positive, while charges can be positive or negative.

$$F = k\frac{q_1 q_2}{r^2} \qquad F = -G\frac{m_1 m_2}{r^2}$$



$$F_{12} = F_{21} = k\frac{q_1 q_2}{r^2}$$

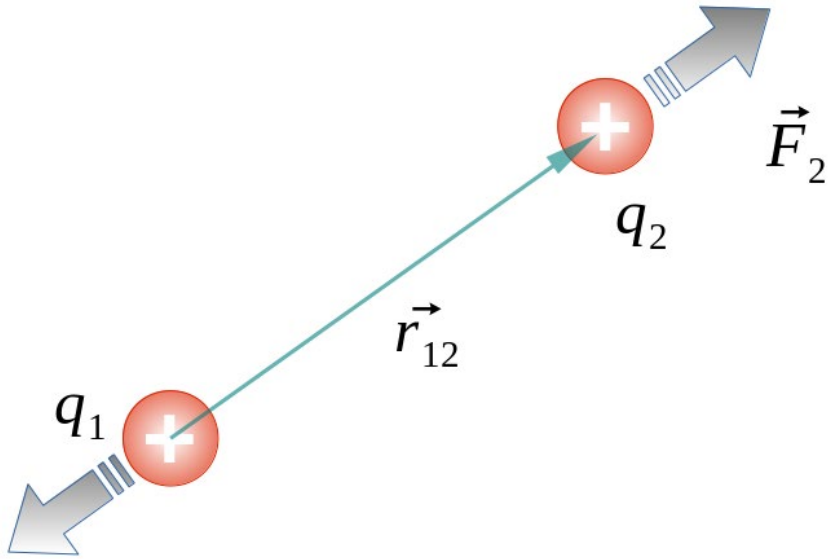
Coulomb law



Forces are vectors: the force between two charges in a 3d space has the same direction of the line connecting the two charges:

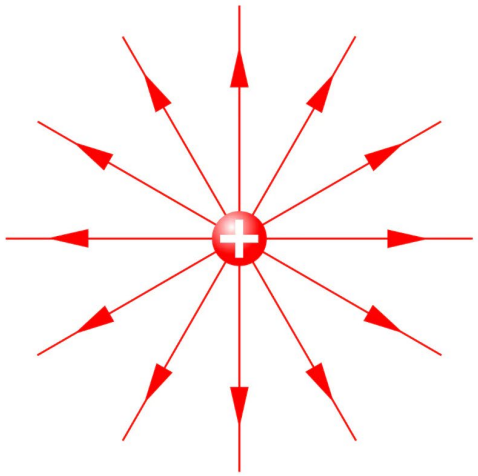
$$\boldsymbol{F} = k\frac{q_1 q_2}{r_{12}^2}\boldsymbol{u}_{r_{12}} = k\frac{q_1 q_2}{r_{12}^3}\boldsymbol{r}_{12}$$

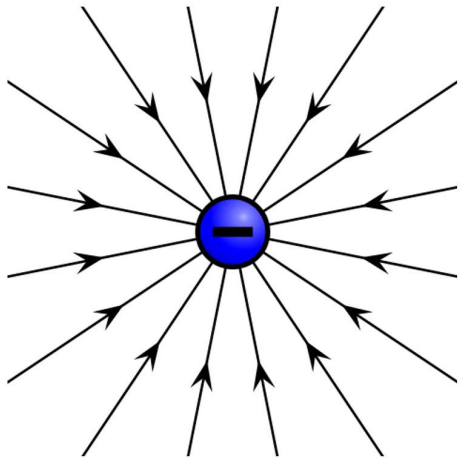
$\boldsymbol{u}_{r_{12}}$ is a unitary vector in the direction of $\boldsymbol{r}_{12}$

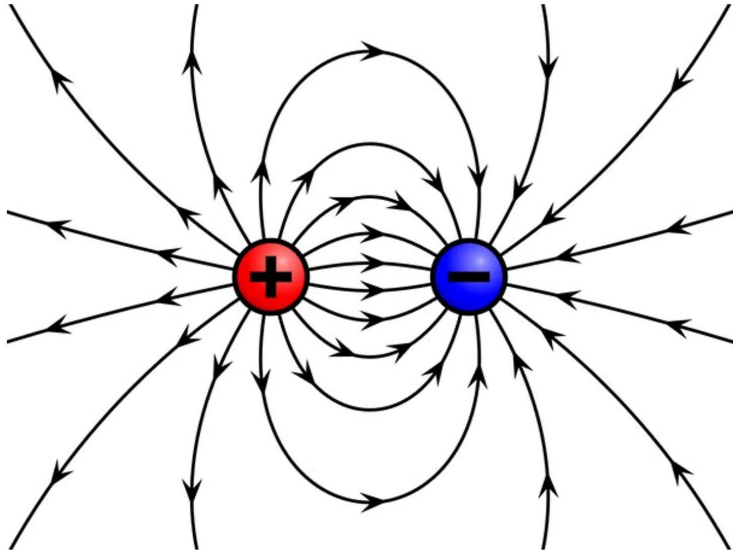


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The electric field







$$\boldsymbol{F} = q \cdot k \frac{Q}{r_{12}^2} \boldsymbol{u}_{r_{12}}$$

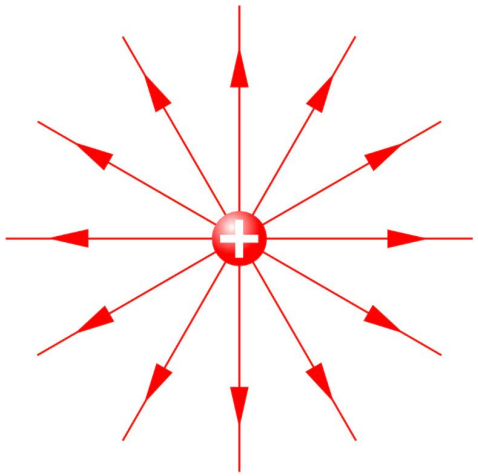
Suppose we keep a charge q fixed in a point in space, and we measure the force $\boldsymbol{F}$ that a second charge q feels, as it moves around the charge Q.

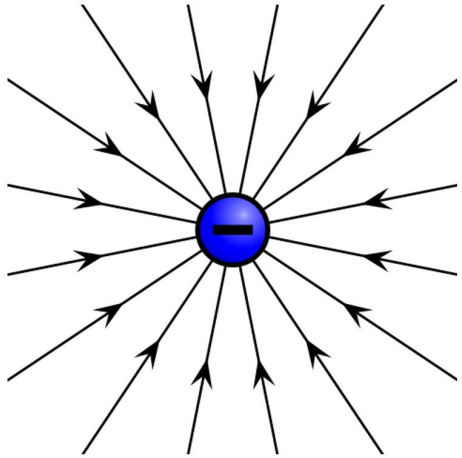
Since the force $\boldsymbol{F}$ is proportional to q, the ratio between $\boldsymbol{F}$ and q_2 depends only on the specific position in space:

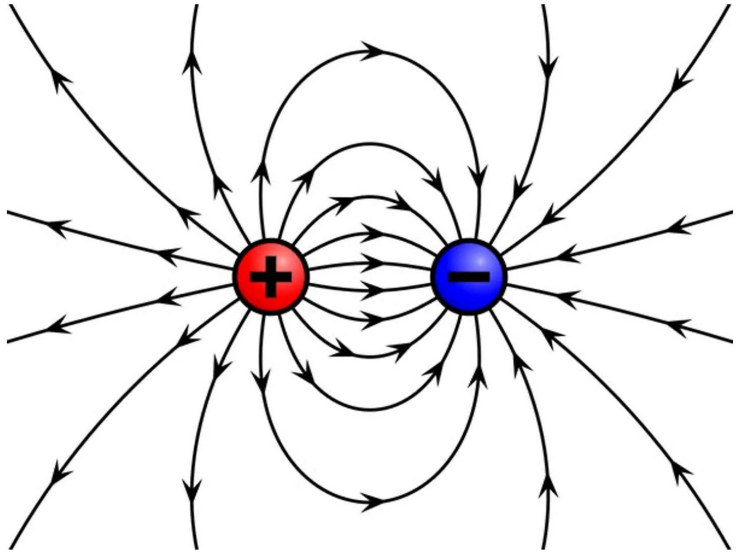
$$\boldsymbol{E}(\boldsymbol{r}) = \frac{\boldsymbol{F}(\boldsymbol{r})}{q}$$

We call $\boldsymbol{E}(\boldsymbol{r})$ the electric field generated by Q and we can think at it as a perturbation of the empty space, due to the existence of Q.

The electric field







The term “field” means that is a function that associate to every point in space a vector:

$$\boldsymbol{E} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$
$$\boldsymbol{r} \mapsto \boldsymbol{E}(\boldsymbol{r})$$

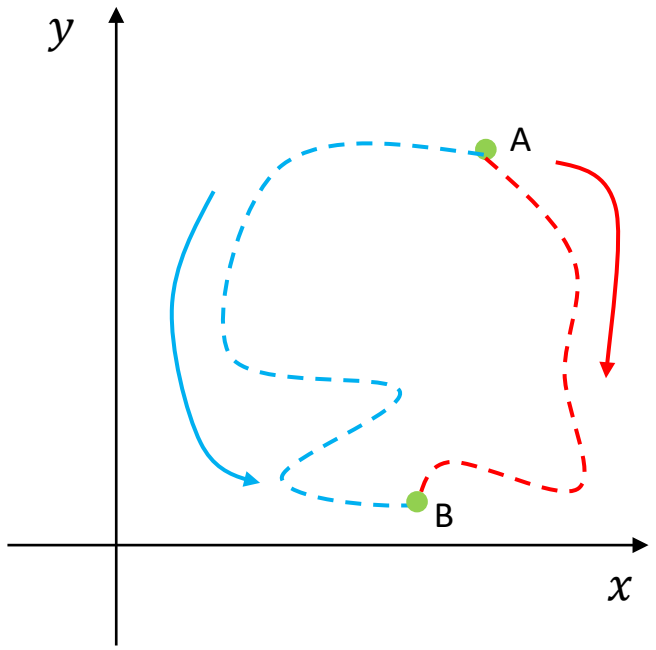
By convention, the electric field “exits” from positive charges and “enters” negative charges.

A positive charge feels a force in the same direction of $\boldsymbol{E}$.

A negative charge feels a force in the opposite direction of $\boldsymbol{E}$.

Electric field generated from different charges can be summed to find the total electric field.

The electrostatic force is conservative



We consider a charged object that moves from point A to point B under the action of an electrostatic force.

It is possible to prove that the work done by the electrostatic force do not depends on the path chosen:

$$W_{AB(R)} = W_{AB(C)}$$

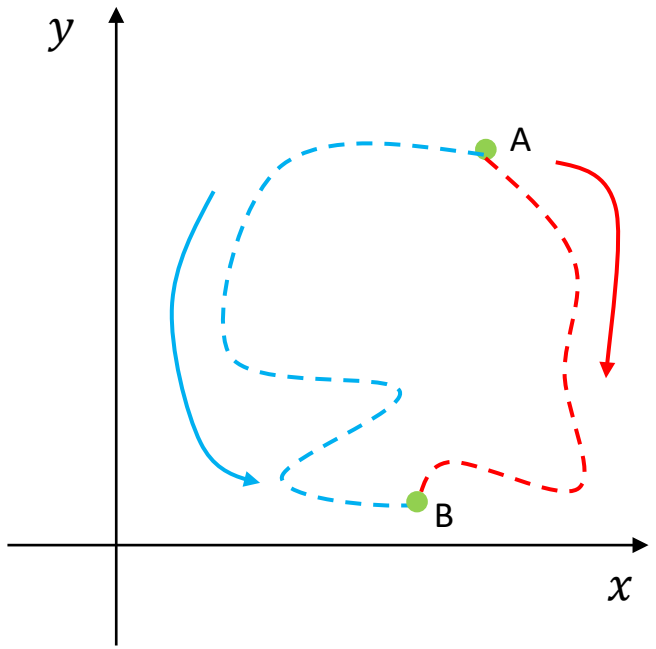
This means that the work done is only a function of the initial and final position.

Therefore, we can write it as the difference of a function which depends on the position:

$$W_{AB} = U_A - U_B = -\Delta U$$

The function U is called the electrostatic potential energy.

Conservation of Energy



The work-energy theorem tells us that the work done to an object by an arbitrary force causes a variation of the kinetic energy.
For conservative forces, this work corresponds to a change in potential energy:

$$W_{AB} = U_A - U_B = K_B - K_A$$

From this equation, we obtain:

$$K_A + U_A = K_B + U_B$$

As A and B are arbitrary points in space, the relationship above is valid for any point in space. If we introduce the quantity $E_{TOT} = K + U$, the total energy of the object we have:

$$E_{TOT} = K + U = const$$

This means that the total energy of an object moving under the effect of an electrostatic field is always conserved.

The force is the gradient of the potential energy

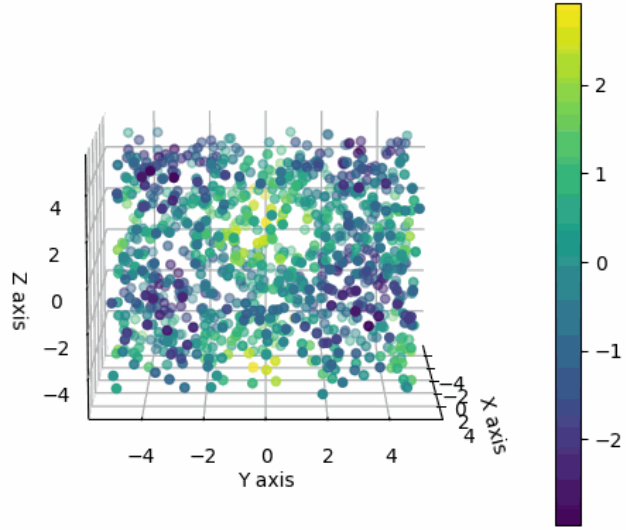
In general, the potential energy does not depend on a single coordinate but is a function of the position in space:

$$U = U(x, y, z)$$

In this case, the force is given by the opposite of the *gradient* of the potential energy:

$$\boldsymbol{F} = -\boldsymbol{\nabla U} = \left(-\frac{\partial U}{\partial x}, -\frac{\partial U}{\partial y}, -\frac{\partial U}{\partial z}\right)$$

In other words, an object experiencing an energy potential, feels a force in the direction in which the energy potential is decreasing most rapidly.

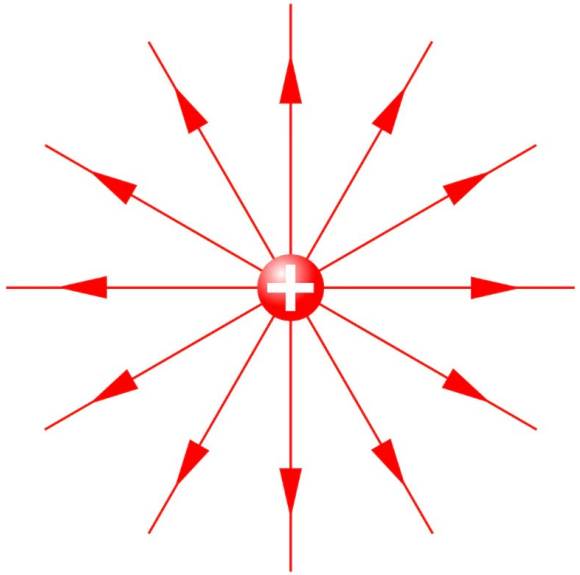


Example of a scalar field in a 3-dimensional space.

The temperature in a room, can have slightly different values depending on the position in space.

Analogously, the potential energy of an object has different values depending on its position in space.

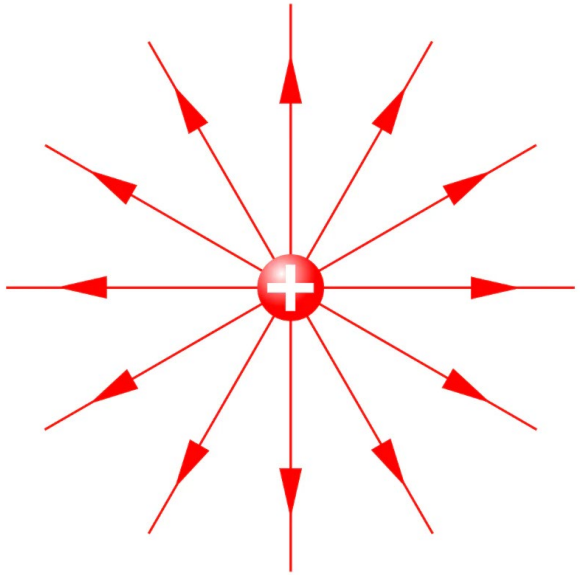
The electrostatic potential

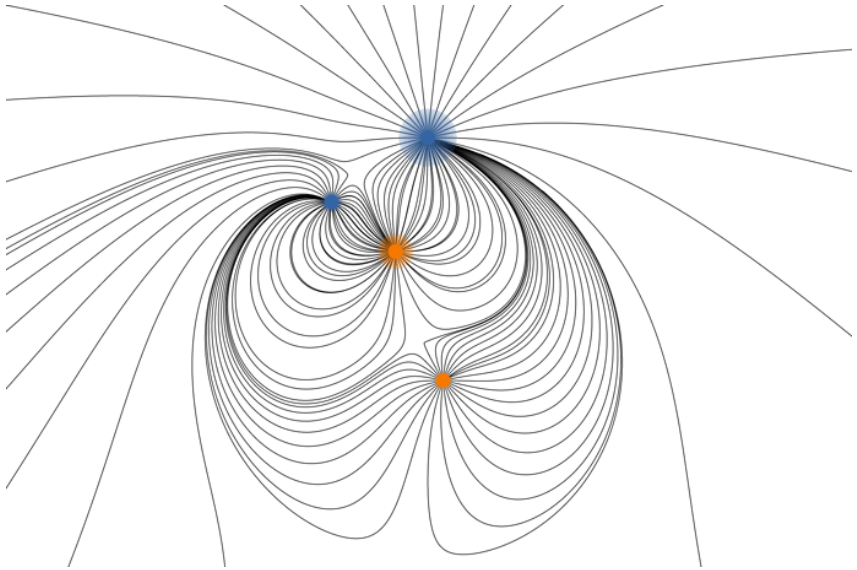


The potential energy of a charge q, which is interacting with another charge Q is proportional to q.

$$\boldsymbol{F} = k\frac{qQ}{r_{12}^2}\boldsymbol{u}_{r_{12}} \Rightarrow U = k\frac{qQ}{r_{12}} = q\left(k\frac{Q}{r_{12}}\right)$$

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This is true in general, the potential energy felt by a charge q is proportional to the charge itself (even if in general the field can be generated by many charges and have a very complicated form).

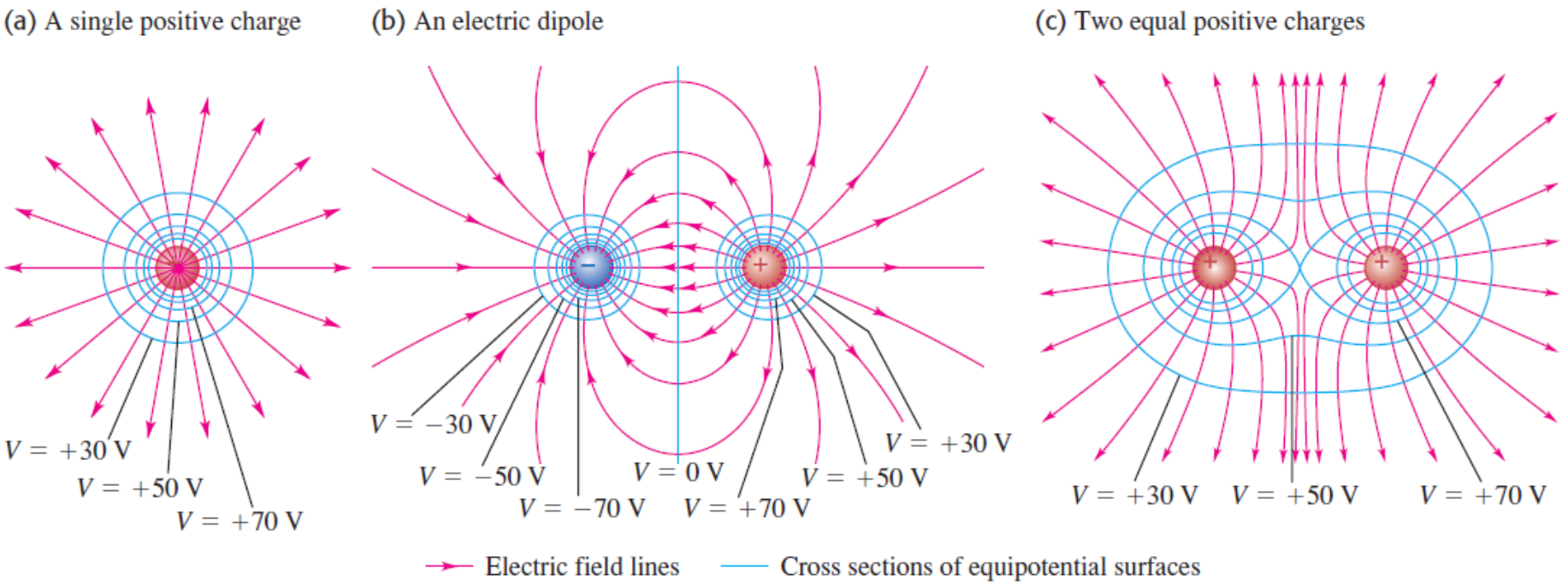
Similarly to the definition of the electric field, we can now define the electrostatic potential as the ration between the electrostatic potential energy and the charge that feels that energy:

$$V(\boldsymbol{r}) = \frac{U(\boldsymbol{r})}{q} \qquad \boldsymbol{E} = -\boldsymbol{\nabla V}$$

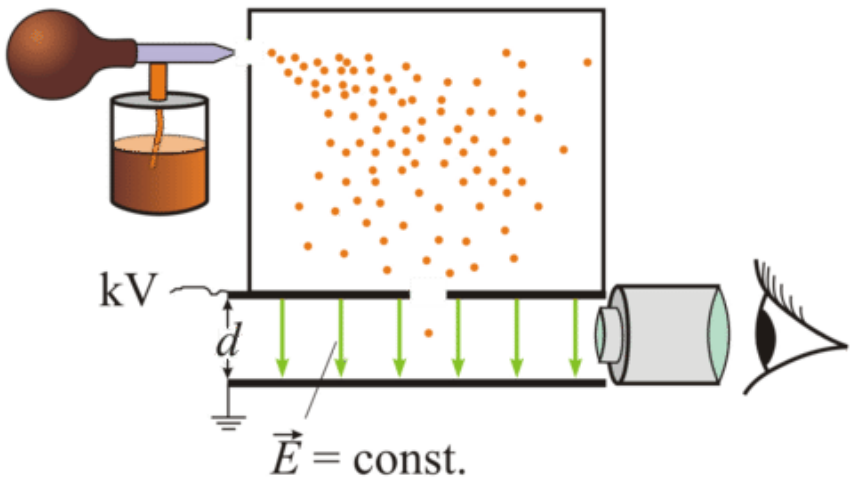
Equipotential surfaces

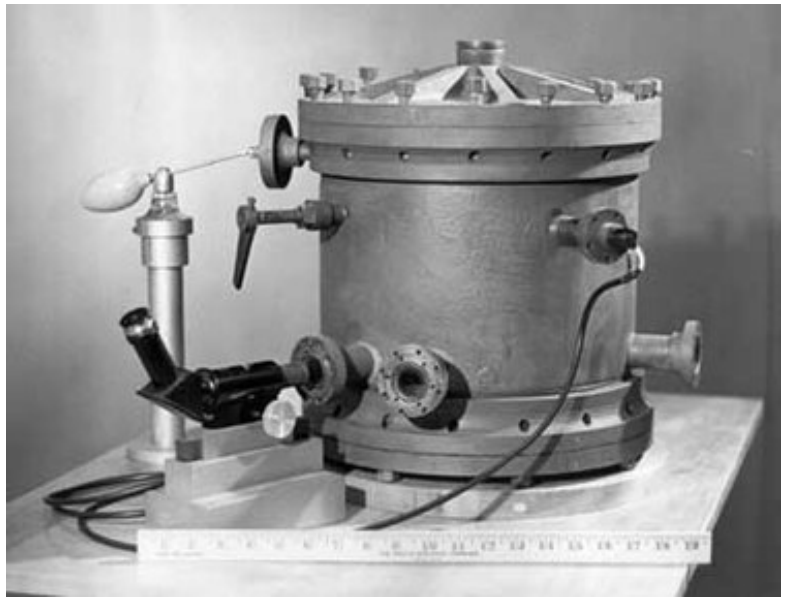
The potential changes along the field lines.
It is useful to consider surfaces formed by points that have the same potential.
These surfaces are called equipotential surfaces.
Electric field lines are always perpendicular to equipotential surfaces.
Equipotential surfaces never intersect (since a point cannot have two different potentials).

23.23 Cross sections of equipotential surfaces (blue lines) and electric field lines (red lines) for assemblies of point charges. There are equal potential differences between adjacent surfaces. Compare these diagrams to those in Fig. 21.28, which showed only the electric field lines.



Millikan's experiment





In a famous experiment carried around 1909, Millikan was able to calculate the charge of the electron.

Droplets of oil were polarized and let fall through a constant electric field.

By changing the electric field, it was possible to equilibrate the force of gravity and obtain estimates of the charge of electron:

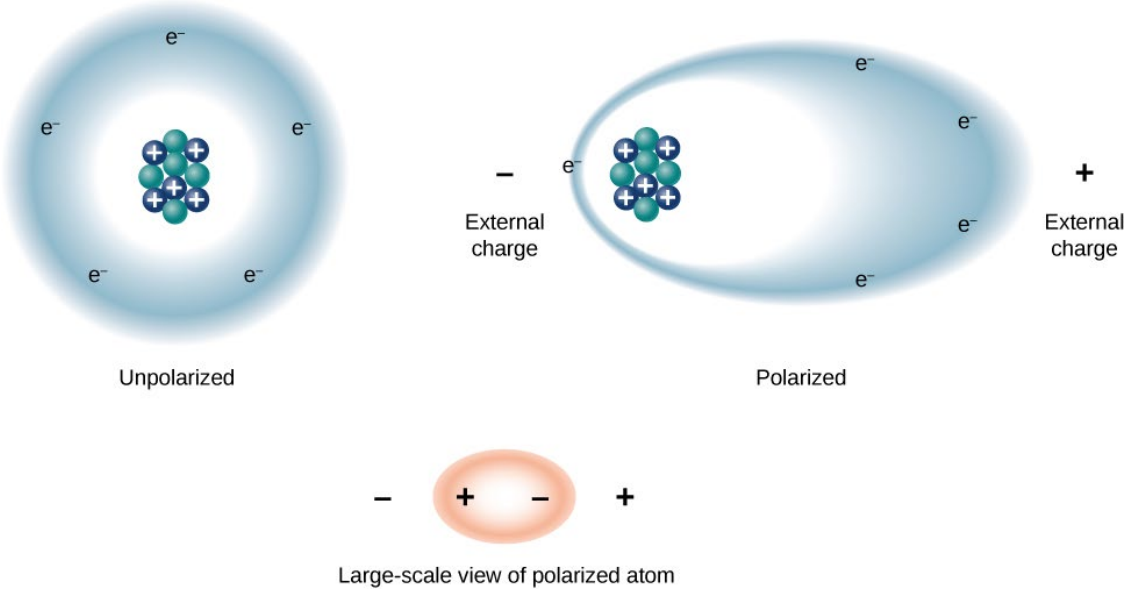
$$qE = mg$$

The mass was estimated from the velocity of the droplets without electric field.

The charge of electron is now estimated at: $-e = -1.6022 \cdot 10^{-19} C$

More importantly, it showed that the charge is quantized, as all the measured charges were integer multiple of e.

Polarization of atoms and molecules

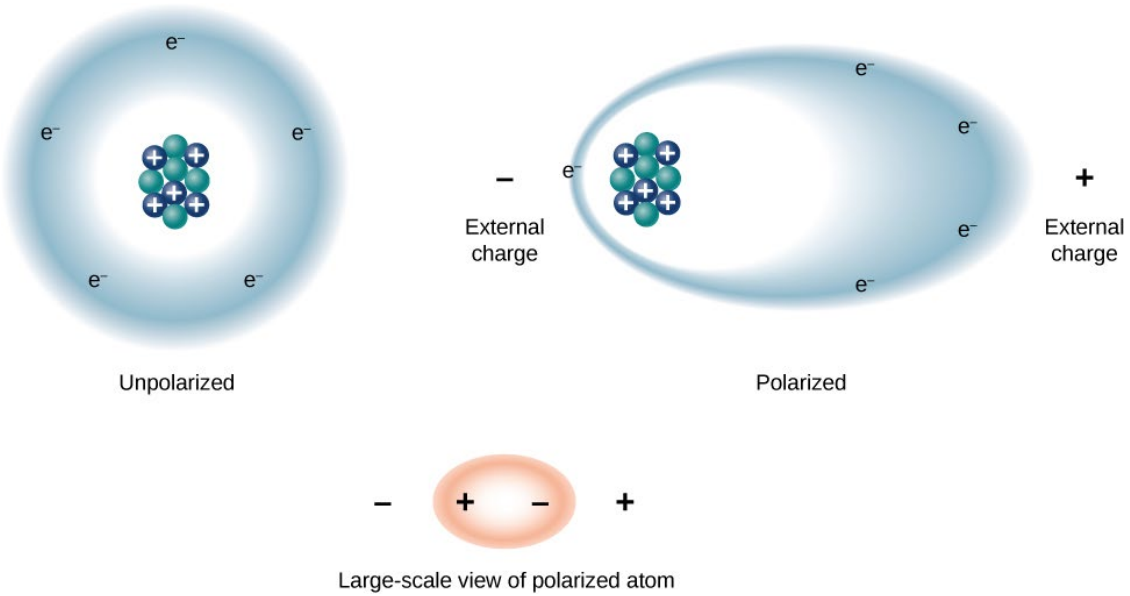


Electrons are distributed in orbitals around the atomic nuclei.

When an external electrostatic field is applied, the electron cloud shifts slightly relative to the nucleus.

This creates an electric dipole moment which tends to align with the external field.

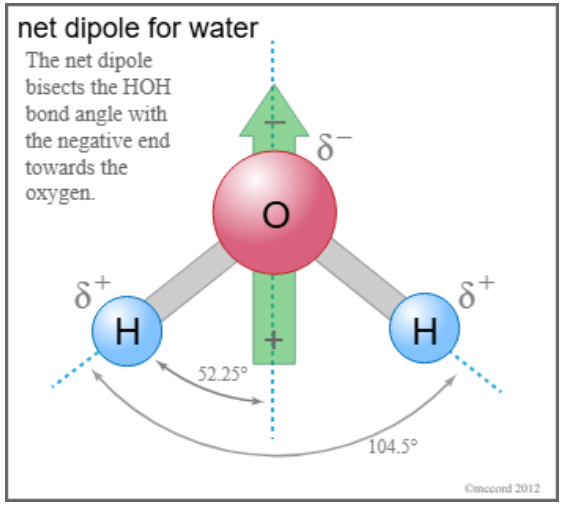
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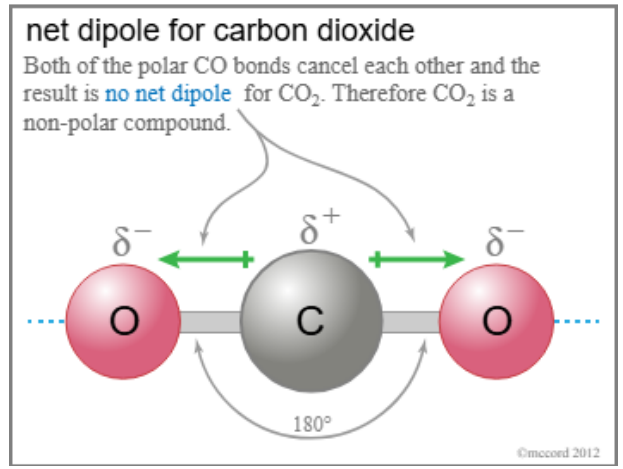
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The electrostatic attraction between electrons and nuclei differs for different atoms.

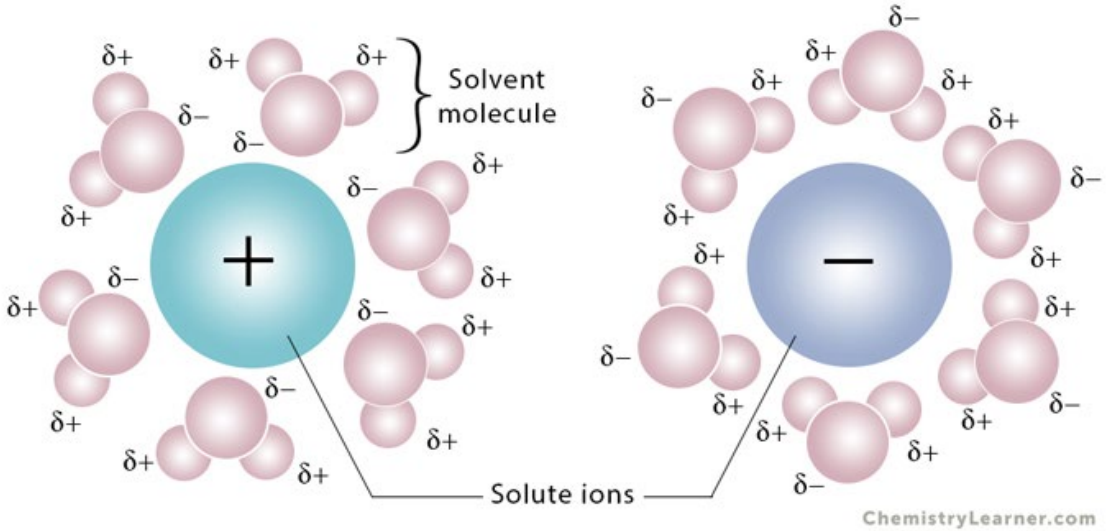
As a result, in a molecule, electrons may be more localized around one atom than another, leading to partial charges.

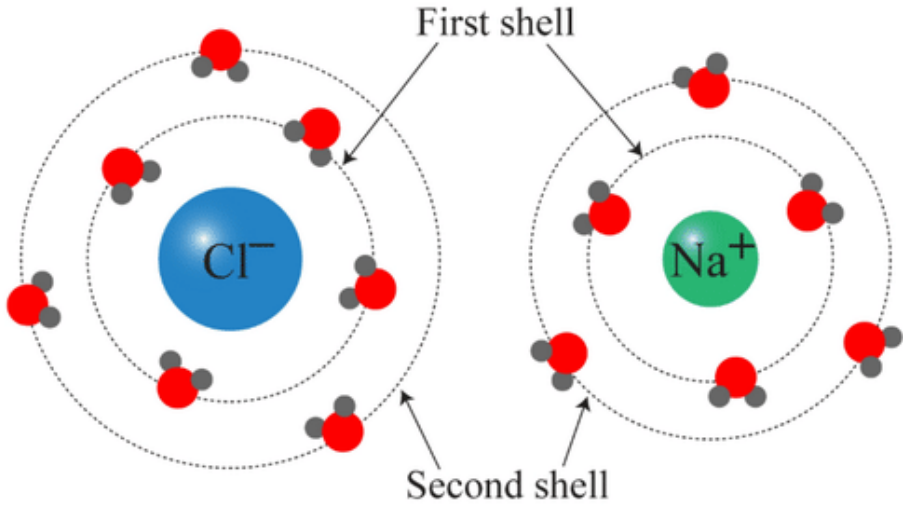
Depending on the geometry, some molecules have a permanent dipole moment, even in the absence of external fields.



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Solvation (hydration) shells





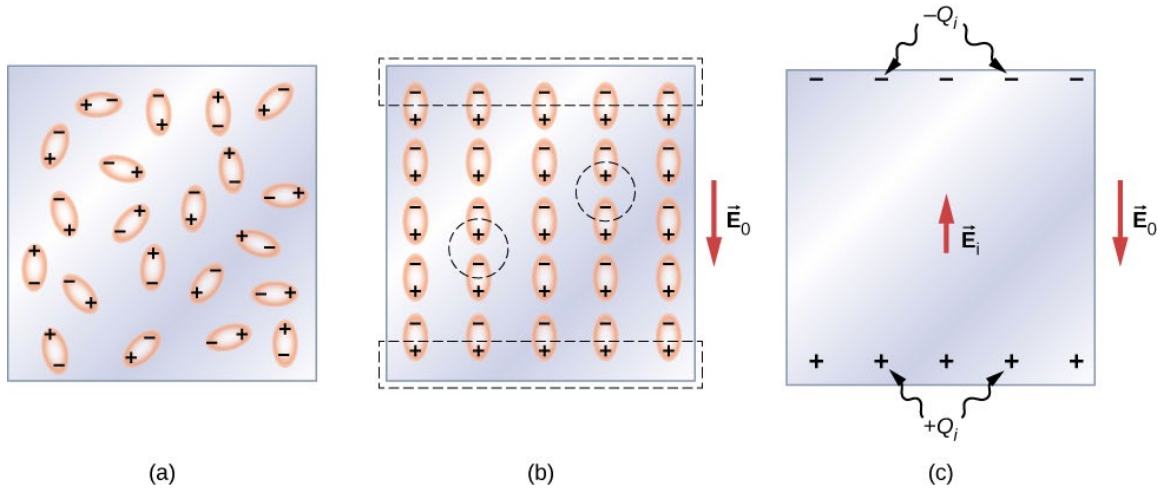
A solvation shell is the structured layer of solvent molecules surrounding a solute (e.g., an ion).

Because water is polar and ions are charged, solvation shells are relatively stable and can mask the ion's charge.

Multiple solvation shells may form, but they become progressively less ordered and less stable farther from the ion.

Solvation shells are dynamic, with solvent molecules constantly exchanging between the shells and the bulk.

Water dielectric constant

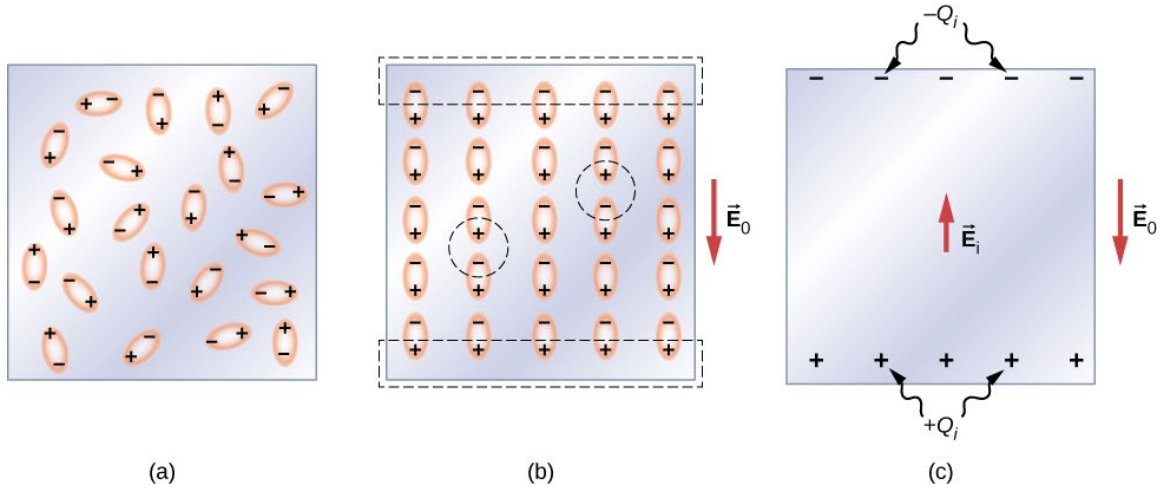


When an external electric field is applied, molecular dipoles (like water) tend to reorient in the direction of the field.

This collective reorientation creates an induced field that opposes the external field, partially canceling it.

The stronger the dipoles and their alignment, the greater the reduction of the effective field inside the medium.

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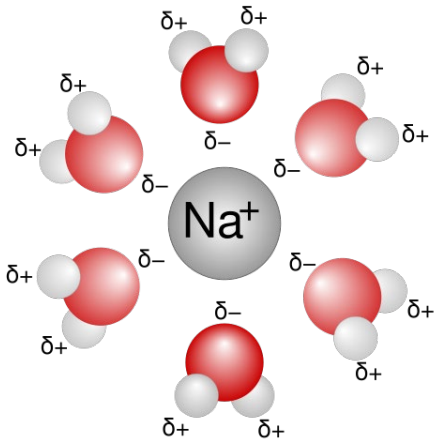
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$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} \rightarrow E = \frac{1}{4\pi\varepsilon_0\varepsilon_r}\frac{q}{r^2}$$

In vacuum

In water



The dielectric constant (ε_r) quantifies this effect: it measures how much a medium reduces the effective electrostatic interaction between charges compared to vacuum.

Water has a high dielectric constant (~80 at room temperature), meaning it strongly screens charges and stabilizes ions and polar molecules in solution.